

RIDDHIMAN MOULICK

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I am a software developer at Oracle, presently working as an applications engineer in the Fusion Release Engineering Team. I have a strong interest in deep learning, computer vision, NLP and algorithmic problem solving, and I am passionate about exploring these fields further, in both academic and professional settings.

EDUCATION

Indian Institute of Technology, Kharagpur <i>CGPA: 9.08/10.00</i>	2020 – 2024
Major: Electrical Engineering Minor: Computer Science & Engineering Micro: Artificial Intelligence	
City International School, Pune <i>Grade: 96.8%</i>	2018 – 2020
Higher Secondary (CBSE)	
The Bishop's School, Undri, Pune <i>Grade: 98.4%</i>	2006 – 2018
Secondary School (ICSE)	

WORK EXPERIENCE

UG Student Researcher CVIR & CNeRG, IIT Kharagpur	Aug' 2023 – May 2024
<i>Transferable Knowledge Systems</i> Prof. Abir Das, Prof. Saptarshi Ghosh (CSE, IIT Kharagpur)	
- Developed an on-the-fly, convolution-based prompting technique coupled with computation of task similarity using language attributes to address catastrophic forgetting in class-incremental continual learning. - Achieved a performance improvement of $\sim 3\%$ over SOTA approaches on benchmark datasets along with significant parameter overhead reduction (Paper accepted at CVPR 2024 Conference). - Designed a fully source-free, actively controlled teacher-update mechanism for continual test-time adaptation (Paper under review at NeurIPS 2024 Conference).	
Cloud Applications Developer Intern Oracle	May 2023 – Jul' 2023
<i>Project Title: Becky Recommendations</i>	
- Implemented the user action recommendation feature in <i>Becky</i>, a digital assistant, to automatically initiate a contextual conversation with users, and suggest actions based on their possible requirements. - Fetched various users' pods and analyzed the data obtained from these, to understand the prospective needs of the user and to alert the user to take necessary actions periodically, via notifications.	

PROJECTS

Computing Game-Theoretic Equilibria <i>Course Project: Algorithmic Game Theory</i>	Fall 2022
- Developed a parser to acquire parameterized inputs in normal form games from the Gambit interface - Computed and enlisted the various strategies for Dominant strategy equilibria and Nash equilibria	
Navigating a Frozen Lake with Stochastic Wind <i>Course Project: Reinforcement Learning</i>	Fall 2023
- Applied Policy Evaluation, Iteration, and Monte-Carlo Algorithms for agent navigation - Visualized the learning iterations and the frozen lake environment using the OpenAI gym package	

COURSEWORK

Computer Architecture and Organisation, Algorithms-I, Deep Learning Foundations and Applications, Reinforcement Learning, Algorithmic Game Theory, AI for Economics, Information Retrieval, Database Management Systems

SKILLS

- Programming Languages:** Python, C, C++, Java, SQL, JavaScript, Verilog, Assembly, $\text{\LaTeX}$
- Tools & Libraries:** PyTorch, Tensorflow, OpenCV, Docker/Podman, Kubernetes, cURL, Matlab, Stateflow

AWARDS AND ACHIEVEMENTS

- Selected to receive the **Chanakya Graduate Fellowship, 2024** by AI4ICPS, IIT Kharagpur
- Achieved an **All India Rank 6** and city rank 1 in ICSE Class X with an aggregate score of **98.4%**
- Awarded the **Silver Certificate** for securing **AIR 233** in Technothon-2016, conducted by IIT Guwahati